

ColorBound: Comparing Menu Dynamics in Virtual Reality

Maxwell Bustamante*, Mike Livingston†, James Hilley‡, Hanniee Tran§, Erik Lovejoy¶, John T. Murray¶

Nicholson School of Communication and Media
University of Central Florida, USA

ABSTRACT

Selection methods in VR are typically pointer-based, with gestural methods including joystick and translation. Rapid selection involves choosing among a small number of options. We propose a novel rotation-based rapid-selection gesture where the angle of the controller is used and present a virtual environment, ColorBound, that compares selection tasks using different control schemes in VR using two scenarios: data visualization and weapon selection. We plan to compare methods through timing and preference. The new scheme, controller rotation shows promise as an alternative with potential applications in tool selection, changing continuous and discrete options, and modifying settings for visualizations.

Keywords: Selection, Manipulation, 3D interaction, UI Control Schemes, VR.

Index Terms: Human-centered computing—Interaction design—Interaction design process and methods—Interface design prototyping;

1 INTRODUCTION

VR applications often require users to make selections in the form of menus. The most typical is a 2D panel that the user points their controller at to select options using a ray visualizer. A typical scenario might have users frequently choosing between a few options (<4-5), and in these cases, a faster method is required. This “quick-select” is common in shooting games where the player must change their weapon during gameplay, often while attempting to navigate enemies or challenges. We present ColorBound: a VR environment that supports comparing quick-selection mechanisms, including a custom method we developed that uses rotation. We validated the usability and suitability of the system through an informal user test, where participants completed the task of selecting (by shooting) colored targets using the appropriate color weapon across all three methods. Our experimental design involves presenting participants with two targets of different colors from the participant’s current gun color. They are then instructed to switch the gun’s color to the color on one of the targets. Participants’ will repeat this for each of the three control schemes listed above. This paper describes the motivation behind focusing on rapid selection, our specific implementation and selected use cases (weapon and graph selection), as well as future work.

2 BACKGROUND

Selection is a fundamental 3D user interaction task. LaViola et al. describe the task as “acquiring or identifying a particular object or

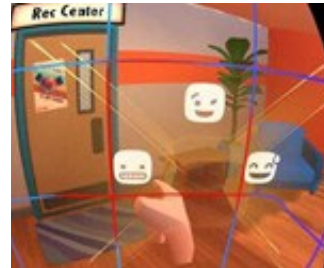


Figure 1: Emoticon menu in Rec Room

subset of objects from the entire set of objects available” [1]. These options can take the form of objects in the Virtual Environment (VE), or 2D panels or elements situated in 3D. Menus are common in both games and applications, with uses ranging from system command and navigation to default interaction behaviors for tools, or even, as in our case, weapons in a shooting game. TiltBrush [2] established a 3D palette as an offhand menu for selecting different brush types, colors, and other tools. It relies on a pointer-based approach to a more traditional menu. Another common scenario is quickly selecting among a small set of options. In these cases, the options are either an assigned button or a gesture as appropriate.

We identified two such “quick-select” gestures: *joystick selection*, where a radial menu is presented and the user makes a selection by pressing the joystick in the direction relative to the center of the displayed menu, and *translation menus*, where the user moves the entire controller in the direction of the desired option, often along a plane perpendicular to the viewer’s gaze. These are used in popular shooting games Space Pirate Simulator [3] and Half Life: Alyx [4]. The VR title Rec Room [5] is a social application where users can trigger temporary changes to their avatar’s emotions through an emoticon menu. This is a good example of where our rotation control scheme would work. The menu uses a significant portion of the player’s view to compensate for the translation selection scheme. A semi-radial rotation control scheme would reduce the amount that the menu occluded the view.



Figure 2: Images of all three control schemes. From right to left: Joystick Selection, Controller Rotation, and Controller Translation

The *joystick scheme* involves selecting from a radial menu that appears when the joystick is pushed in a direction. Its direct analogue is Space Pirate Trainer [3]. Our implementation of the joystick scheme differs in that it requires a button press. In Space Pirate Trainer, the player only has to flick the joystick in a direction to switch their weapon type. The *translation scheme* involves players physically moving their hand in one of four directions to make a selection. Its direct analogue is Half Life: Alyx [4], which

* maxwellbustamante00@gmail.com

† r.mike.livingston@gmail.com

‡ j_hilley28@yahoo.com

§ hannieetran42@knights.ucf.edu

¶ elovejoy27@knights.ucf.edu

¶ jtm@ucf.edu

uses a similar method to switch weapons. The *rotation scheme* involves the player rotating their wrist to choose an option from a semi-radial menu, which has no direct analogue that we found.

Our rotation scheme is implemented using Unity and the XR Interaction Toolkit. Each controller's rotation is tracked relative to the players' controller position. The radial menu display's selector is a child of the controller object. Then, an angle is calculated between the horizon and the menu selector, and the current selection is calculated based on that angle. Selection for rotation is dependent on the controller axis; pitching the controller forward just causes the axis that is tracked for selection to follow. This is based on how OpenXR treats controllers, since this technique simply tracks the position of an empty game object, which is made to be a child to the controller. The axis location varies from controller to controller, but generally extends from the controller in the direction of the player's hand.

For translation, the axis starts centered on the controller, and gradually orients itself to face the HMD. On the axis drawn directly from the HMD to the menu, no selections are made. However, the controller's position is still tracked along it. If the controller moves a certain distance along this HMD-controller axis away from the menu in either direction, the system assumes that the player is trying to get a better look at the menu. It will then re-center the menu to wherever the player's hand is now. This becomes the new menu origin, and translation selection tracking continues normally.

3 PLANNED EVALUATION AND ITERATIONS

ColorBound presents three selection mechanisms in a shooting game scenario. We plan to evaluate through an experiment each gestural scheme: *joystick*, *translation*, and *rotation*. We plan to evaluate player preference as well as effectiveness. During each of the three conditions, participants shoot virtual targets with the appropriate weapon. These targets are color-coded with one of four different colors. Players can only successfully clear the targets by changing their held gun to one of the same colors as the target. Players will perform the selection and targeting task for each of the three control schemes in a random order. The player's performance will be measured in the time it takes to clear targets as well as their speed at changing to the correct colors. Player's accuracy in hitting the targets is not a factor for the recorded data, because the targets are large and spawn in the same place every round.

We hypothesize that using controller rotation would be preferred to either joystick-based selection or translation-based selection in a quick-select scenario involving color targets. We believe that moving the wrist within its range of motion is both intuitive and comfortable, but that it won't be the fastest of the three.

One change we made from the initial design was to have the gun switch to a neutral color after shooting. This avoids having the color of the gun match the targets at the onset of the task. Another change we made was to give participants the ability to test out the control schemes before moving on to the trial. This will help participants get comfortable with the control schemes before using them in recorded data. We reduced users to one controller. To combat issues that sprung up during development while using both controllers, we only allowed the participants to use one. Another iteration we made is to require each participant to demonstrate understanding of how each control scheme works before starting the trial. We discovered that an empty play space was not adequate for participants to want to use the control schemes beforehand.

4 ROTATIONAL APPLICATION BEYOND COLORBOUND

The rotational modal selection can easily be applied to many other uses within a virtual setting. For example, users may want to

modify terms and filters, such as switching between graphs or information quickly, for which radial rotation-based selection is a previously unexplored option. In our second application scenario, we re-utilized the selection schemes to filter data and information. This situation uses each of the selection schemes to switch between four different metrics related to the COVID-19 pandemic: new cases, deaths, vaccinations, and hospitalizations. This information was found using Google's COVID-19 Open Data Repository [6]. This setting shows that the rotation selection method is a competitive option for visualizing and manipulating UI in VR, since it can easily apply to several scenarios in a way that is just as, or more convenient than the other two methods.

Rotational selection is applicable elsewhere. For instance, in an art application, users will choose among colors for objects. Similarly, context menus could also be represented as rotational menus. Probably the best use is to offer a rapid selection of different "tools" as is offered in Rec Room's Maker Pen, which currently uses a flat menu. In these cases, it may be useful to represent the options in a hierarchical list. Rotation can be applied to continuous options, such as volume or brightness of a game in a settings menu, adjusting numbers in a numerical input field, or scrolling along a timeline (with the rotation angle representing the rate).

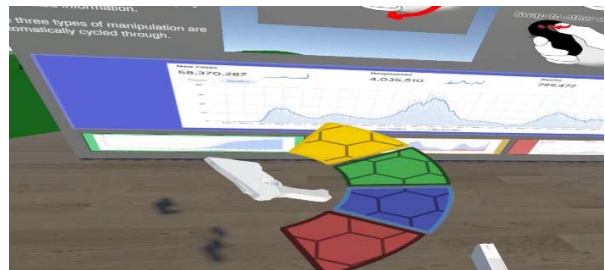


Figure 3: The control schemes used in a graph selection scenario. The semi-radial rotational control scheme is shown here.

From our experience so far, the rotation control scheme should be limited to 2-3 options. Future work in comparing gestural menu options in VR include trigger criteria, such as button pressing or releasing, and sensitivity to threshold values (e.g. dead zones). Combining input from both controllers is another promising direction, but we predict that adding additional modalities (e.g. combining translation and rotation) may overwhelm users over sequences of simpler methods for hierarchical menus.

We believe rotational gesture presented here represents a viable alternative gesture menu for quick-select use cases in 3D VR applications.

REFERENCES

- [1] J. Jr. Laviola, E. Kruijff, R. P. McMahan, D. Bowman, and I. P. Poupyrev, *3D User Interfaces: Theory and Practice*, 2 edition. Boston: Addison-Wesley Professional, 2017.
- [2] *Tilt Brush*. Google, 2016.
- [3] *Space Pirate Trainer*. Geraardsbergen, Belgium: I-Illusions, 2017.
- [4] *Half Life: Alyx*. Bellevue, Washington, USA: Valve, 2020.
- [5] *Rec Room*. Seattle, Washington, USA.
- [6] O. Wahltinez and others, "COVID-19 Open-Data: curating a fine-grained, global-scale data repository for SARS-CoV-2," 2020, [Online]. Available: <https://goo.gle/covid-19-open-data>